

By defn. of AND, $((a_1=a_1) \vee (a_1=a_2)) \wedge ((a_2=a_1) \vee (a_2=a_2))$ is True
 and $((a_1=a_2) \vee (a_1=a_1)) \wedge ((a_2=a_2) \vee (a_2=a_1))$ is True.

By defn. of conditional, $((a_1 \neq a_2) \rightarrow ((a_1=a_1) \vee (a_1=a_2)) \wedge ((a_2=a_1) \vee (a_2=a_2)))$
 and $((a_2 \neq a_1) \rightarrow ((a_1=a_2) \vee (a_1=a_1)) \wedge ((a_2=a_2) \vee (a_2=a_1)))$ are both True
 respectively.

By defn. of AND, $((a_1 \neq a_1) \rightarrow ((a_1=a_1) \vee (a_1=a_1)) \wedge ((a_2=a_1) \vee (a_2=a_1))) \wedge$
 $((a_1 \neq a_2) \rightarrow ((a_1=a_1) \vee (a_1=a_2)) \wedge ((a_2=a_1) \vee (a_2=a_2))) \wedge$
 $((a_2 \neq a_1) \rightarrow ((a_1=a_2) \vee (a_1=a_1)) \wedge ((a_2=a_2) \vee (a_2=a_1))) \wedge$
 $((a_2 \neq a_2) \rightarrow ((a_1=a_2) \vee (a_1=a_2)) \wedge ((a_2=a_2) \vee (a_2=a_2)))$ is True.

$\therefore$ If common domain is of at most 2 elements, then $\forall x \forall y ((x \neq y) \rightarrow$
 $\forall z ((z=x) \vee (z=y)))$ is True. (Direction 1)

Direction 2: $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z=x) \vee (z=y)))$ is True $\Rightarrow$ Common
 domain consists of at most 2 elements

Proof by contraposition

Let common domain consists of ~~at least~~ at least 3 elements a_1, a_2 and a_3 .
 $(a_1 \neq a_2)$ is True. $(a_3=a_1)$ is False and $(a_3=a_2)$ is False. By defn. of
 OR, $(a_3=a_1) \vee (a_3=a_2)$ is False. By defn. of universal quantifier,
 $\forall z ((z=a_1) \vee (z=a_2))$ is False. By defn. of conditional, $(a_1 \neq a_2) \rightarrow \forall z$
 $((z=a_1) \vee (z=a_2))$ is False. By defn. of universal quantifier,
 $\forall y ((a_1 \neq y) \rightarrow \forall z ((z=a_1) \vee (z=y)))$ is False. By defn. of universal
 quantifier, $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z=x) \vee (z=y)))$ is False. $\square$ which

35) Find a common domain for the variables x, y, z and w for the
 statement $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is true and another
 common domain for these variables for which it is false.